xpath locator

XPath stands for XML Path Language. It is a way to find and select a specific element on a web page by describing its location, like an address.
XPath is multi-directional.

Types of XPath

Absolute XPath
1)Starts from the root and follows the full path, beginning with a single /

```
<html>
 <body>
   <div>
     <form>
       <input type="text">
     </form>
   </div>
 </body>
</html>
```

2)Example: /html/body/div/form/input
3)Easy to write, but it breaks if the page structure changes even slightly.
4)It is long/complex and hard to read.

Relative XPath
1)Starts anywhere in the document, beginning with //.

```
<html>
 <body>
   <div>
     <form>
       <input id='email' type="text">
     </form>
   </div>
 </body>
</html>
```

2)Example: //input
3)More flexible and reliable, so it's the one used most in practice.

Categories of Relative XPath

1)xpath By attribute
2)xpath By text()
3)xpath By contains
4)xpath By multiple attributes
5)xpath By surrounding
6)xpath By starts-with
7)xpath By ends-with
8)xpath By index
9)xpath By normalize-space
10)xpath By axes

xpath by Attribute

locating an element on a web page by using the value of one of its attributes, such as id, name, or class etc.

Syntax:

```
//tagname[@attributeName='Attributevalue']
```

xpath by text()

locating an element on a web page by using the visible text written inside its tag.

Syntax:

```
//tagname[text()='visible text']
```

xpath by contains

locating an element on a web page by passing only a part/partials amount of its textvalue or attribute value, instead of the full value.
It is helpful when the full text or attribute value is long/lengthly or changes dynamically, but one part of it stays the same.

Syntax:

```
//tag[contains(text(),'partial text')]

//tag[contains(@attribute,'partial value')]
```

xpath By multiple attribute

XPath by multiple attributes is used to locate a web element uniquely on the web page by combining two or more attribute-value pairs with the logical operators and / or, when a single attribute-value pair is not unique.

Syntax:

```
//tag[@attribute1='value1' and @attribute2='value2']

//tag[text()='textvalue' and @attribute='value']

//tag[@attribute1='value1' or @attribute2='value2']
```

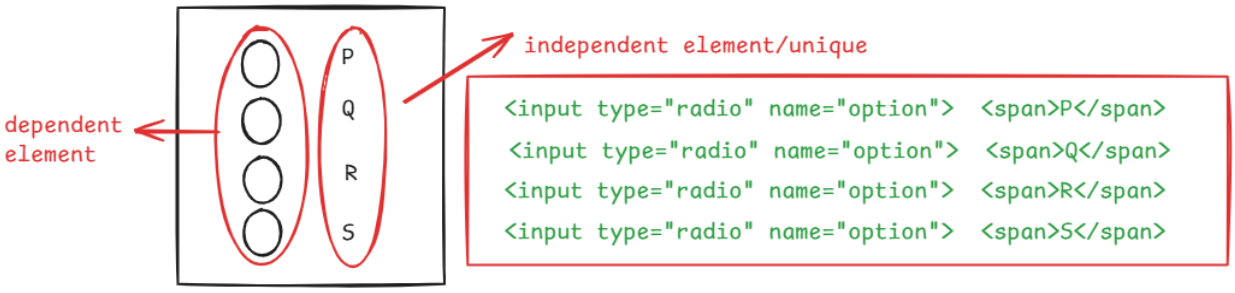
xpath By surrounding

locate a web element uniquely on a web page, when several elements share the same tag name and the same attribute-value pair, by starting from a nearby unique element and moving to the target element is called as xpath by surrounding
it is also called as independent and dependent xpath
1)independent = unique element
2)dependent=common element

To locate dependent and independent elements, we need to follow 4 steps:

1)Identify which is dependent and which is independent.
2)Write the XPath for the independent element first.
3)Traverse back to the immediate common parent.
  How to traverse back: Using /..
  (We have to traverse back until both the independent and dependent elements are getting highlighted.)
4)Update the XPath expression by writing the XPath for the dependent element.

for eg:



Problem: In this I want to loacte 3 radio button
If we write //input[@name='option'], it matches all 4 radio buttons.

Solution, using the 4 steps:

Identify: the radio button is dependent, the text "R" is independent.
Independent XPath: //span[text()='R']
Traverse back to the common parent: //span[text()='R']/..
Dependent XPath: //span[text()='R']/../input

xpath By starts-with

locating a web element on a web page by passing the beginning of its text or attribute value, instead of the full value.

Syntax:

```
//tag[starts-with(@attribute,'starting value')]

//tag[starts-with(text(),'starting text')]
```

xpath by index

locating a web element on a web page by using its position number when several elements match the same XPath.
In XPath, indexing starts from 1, not 0.

Note: Indexing is not recommended, because if the page changes (an element is added or removed), the position changes and the XPath breaks. Use it only when the web application is static; otherwise prefer XPath by surroundings.

Syntax:

```
(//tag[@attribute='value'])[index]
```

XPath index: last() and position() methods

1.last()=last() is a method that selects the last element among the matching elements.

Syntax:

```
(//tag[@attribute='value'])[last()]
```

2. position()=position() is a method that selects an element by its position number among the matching elements and locate multiple web elements at a time based on a condition.

Syntax:

```
(//tag[@attribute='value'])[position()=number]
```

xpath By normalize-space

locating a web element by passing its visible text or attribute value after the leading and trailing spaces are removed/ignored, which is used when an XPath written with the text() function fails unexpectedly because of whitespace.

Notes:

1)If an XPath using the text() function fails unexpectedly, or the attribute-value pair contains extra spaces, the first suspect is a whitespace issue, and normalize-space() can be used to fix it.

2)If the visible text or attribute value has leading or trailing spaces, a normal XPath will not locate that web element.

3)normalize-space() ignores the leading and trailing spaces while locating the web element.

Syntax:

```
//tag[normalize-space()='text']

//tag[normalize-space(@attribute)='value']
```

or

```
//tag[normalize-space(text())='text']

 //tag[normalize-space(@attribute)='value']
```

TS xpath.spec.ts

tests > TS xpath.spec.ts > ...

```ts
import {test} from "@playwright/test"
Run | Debug | Inspect
test("xpathbyattribute",async({page})=>{
await page.goto("https://opensource-demo.orangehrmlive.com/web/index.php/auth/login")
await page.locator("//input[@name='username']").fill("Admin")
await page.locator("//input[@name='password']").fill("admin123")
await page.locator("//button[@type='submit']").click()
await page.waitForTimeout(2000)
})
Run | Debug | Inspect
test("xpathbytext",async({page})=>{
await page.goto("https://www.saucedemo.com/")
await page.locator("#user-name").fill("standard_user")
await page.locator("[name='password']").fill("secret_sauce")
await page.locator(".submit-button").click()
await page.locator("//div[text()='Sauce Labs Backpack']").click()
await page.waitForTimeout(2000)
})
Run | Debug | Inspect
test("xpathbycontains",async({page})=>{
await page.goto("https://www.amazon.in/")
await page.locator("//input[contains(@placeholder,'Amazon')]").fill("iphone")
await page.locator("//input[contains(@id,'submit-button')]").click()
await page.locator("//span[contains(text(),'iPhone 16 128 GB: 5G Mobile Phone with Camera Control')]").first().click()
await page.waitForTimeout(2000)
})
```

TS xpath.spec.ts

tests > TS xpath.spec.ts > ...

```ts
Run | Debug | Inspect
test("XpathByMultipleAttribute",async({page})=>{
await page.goto("https://en.wikipedia.org/wiki/Main_Page")
await page.locator("//input[@type='search' and @name='search' and @title='Search Wikipedia [alt-f]']").fill("hello")
await page.waitForTimeout(2000)
})

Run | Debug | Inspect
test("XpathBySurrounding",async({page})=>{
 await page.goto("https://demowebshop.tricentis.com/gift-cards")
 await page.
 locator("//a[text()='$25 Virtual Gift Card']/../..//input[@value='Add to cart']").click()
await page.waitForTimeout(2000)
})
Run | Debug | Inspect
test("XpathByStarts-with",async({page})=>{
 await page.goto("https://www.amazon.in/")
await page.locator("//input[starts-with(@id,'twotab')]").fill("mobile")
await page.locator("//input[starts-with(@id,'nav')]").click()
await page.locator("//span[starts-with(text(),'Samsung Galaxy S25 Ultra 5G')] ")
.first().click()
await page.waitForTimeout(2000)
})
Run | Debug | Inspect
test("XpathByindex",async({page})=>{
await page.goto("https://demowebshop.tricentis.com/")
// await page.locator("(//input[@value='Add to cart'])[5]").click()
// await page.locator("(//input[@value='Add to cart'])[last()]").click()
// await page.locator("(//input[@value='Add to cart'])[position()=4]").click()
await page.locator("(//a[normalize-space(text())='Books'])[1]").click()
await page.waitForTimeout(2000)
})
```